

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	45
2	EST	55

UCS553 : ENTERPRISE WEB APPLICATION

L	T	P	Cr
3	0	2	4.0

Course Objectives: To equip undergraduate students with foundational and intermediate concepts of Java programming, object-oriented design, and hands-on coding. By the end of the course, students will be able to build a mini-project in Java that exhibits OOP concepts, exception handling, file operations, and collection usage.

Unit 1: Introduction to Java Programming: History and evolution of Java; Features of Java (platform independence, security, portability); JVM, JDK, JRE – architecture and workflow; Compilation & execution process; Structure of a Java program; Data types, variables, type casting; Operators and expressions; Input/output (Scanner, command-line arguments)

Unit 2: Control Statements & Arrays: Conditional statements (if, if-else, switch); Loops (for, while, do-while); Break, continue, labelled loops; Arrays (1D, 2D); Array operations and common algorithms; Intro to Strings and StringBuffer

Unit 3: Object-Oriented Programming in Java: Class and Object creation; Constructors & constructor overloading; Static members and static blocks; Encapsulation & data hiding; Inheritance (single, multilevel, hierarchical); Method overloading & overriding; Super and this keyword; Final, abstract classes, and interfaces; Polymorphism (compile-time & runtime)

Unit 4: Exception Handling & Packages: Types of errors: compile-time vs runtime; Exception hierarchy; try-catch-finally; throws & throw; Custom exceptions; Creating and using packages; Access modifiers (public, private, protected, default)

Unit 5: File Handling & Java Collections: File I/O using FileInputStream, FileOutputStream, FileReader, FileWriter; Serialization & deserialization; Collections framework overview; List, Set, Map interfaces; ArrayList, LinkedList, HashMap, TreeSet; Iterators & enhanced for-loop; Wrapper classes & autoboxing/unboxing

Unit 6: Multithreading & Basic GUI (Optional but Recommended): Thread lifecycle; Creating threads: Thread class & Runnable interface; Synchronization basics; Inter-thread communication basics; Introduction to Java GUI (AWT/Swing); Building simple forms: buttons, labels, text fields; Event handling model

Unit 7: Project Development Module: Requirement analysis & problem definition; Designing class diagrams (basic UML); Modular programming practices; Integrating multiple Java concepts; Code testing, debugging & documentation; Final mini project demonstration

Course Learning Outcomes (CLOs) /Course Objectives (COs):

After completing this course, students will be able to:

1. Understand JVM architecture, data types, control constructs, and OOP concepts.
2. Implement classes, objects, inheritance, interfaces, and polymorphism.
3. Apply exception handling, file I/O, collections, and Java utilities.
4. Develop simple GUI/console-based applications.
5. Build and demonstrate a functional **mini-project** using Core Java.

Textbooks & References

- Herbert Schildt, *Java: The Complete Reference*, McGraw Hill
- E. Balaguruswamy, *Programming with Java*
- Kathy Sierra & Bert Bates, *Head First Java*

SUGGESTED MINI PROJECTS (Core Java Level)

Console-Based Projects

1. **Student Record Management System**
2. **Library Management System**
3. **Employee Payroll Management**
4. **Bank Account Simulation (ATM-like menu)**
5. **Car Rental Booking System**
6. **Course Registration System**
7. **Bill Management System (electricity/water)**
8. **Inventory Management System**

GUI-Based Projects (Swing/AWT)

1. **Simple Calculator**
2. **Notepad Text Editor**
3. **Personal Expense Tracker**
4. **Quiz Application**
5. **Tic-Tac-Toe Game**
6. **Basic Chat Simulation (offline, using threads)**
7. **Hotel Room Booking System**

Projects Involving File Handling

1. **Daily Logbook / Diary application**
2. **Password Manager (file-based encryption)**
3. **Data Serialization-Based Student Report System**